

Mozes Jacobs

425-273-5242 | mozesjacobs@g.harvard.edu | mozesjacobs.github.io

EDUCATION

Harvard University

Ph.D. in Computer Science | Advisor: Demba Ba | Kempner Graduate Fellow
Computer Vision, Representation Learning, Interpretability, Foundation Models

Cambridge, MA
2023 – 2028 (Expected)

University of Washington

B.S. in Computer Science | Summa Cum Laude | GPA: 3.99 / 4.00

Seattle, WA
2018 – 2022

PUBLICATIONS & PREPRINTS

- [1] **A Phenomenology of Attention Sinks: Two Algorithms, Two Solutions** Under Review
Fesser L.*, **Jacobs M.***, Fel T.*, Keller TA., Kakade S.
- [2] **Block-Recurrent Dynamics in ViTs** ICLR 2026
Jacobs M.*, Fel T.*, Hakim R.*, Brondetta A., Ba D., Keller TA.
- [3] **Traveling Waves Integrate Spatial Information Through Time** CCN 2025 (oral)
Jacobs M., Budzinski RC., Muller L., Ba D., Keller TA.
- [4] **HyperSINDY: Deep Generative Modeling of Nonlinear Stochastic Governing Equations** Preprint 2023
Jacobs M., Brunton BW., Brunton SL., Kutz JN., Raut

RESEARCH EXPERIENCE

Kempner Institute | Harvard University | Graduate Research Fellow

2023 – Present

- **Block-Recurrent Foundation Models:** Showed foundation models (e.g., DINOv2) can be distilled into recurrent surrogates that recover **96%** of ImageNet-1k accuracy with only **2 recurrent blocks**, enabling principled model compression.
- **Dynamical Interpretability:** Characterized ViT depth as a dynamical flow into low-dimensional attractors, revealing token-specific self-correcting trajectories and low-rank late-layer updates.
- **Neuro-Inspired Architectures:** Designed convolutional recurrent models using traveling waves for spatial integration, outperforming U-Nets on semantic segmentation.

Allen Institute | AI Institute in Dynamic Systems | Shanahan Foundation Predoctoral Fellow

2022 – 2023

- **HyperSINDy:** Built a deep generative framework combining SINDy with amortized variational inference for uncertainty-aware discovery of stochastic governing equations. (*Advisors: R. Raut, J.N. Kutz, S. Brunton*)

Neural Systems Lab | University of Washington | Undergraduate Research Assistant

2019 – 2022

- **Predictive Coding:** Proposed *Gradient Origin Predictive Coding*, a video generation framework unifying cortical predictive coding with Bayesian deep learning for next-frame prediction. (*Advisor: Rajesh Rao*)

Noble Lab | University of Washington | Undergraduate Research Assistant

2019 – 2022

- **Genomic Algorithms:** Optimized *PASTIS*, a Poisson-based 3D chromatin structure inference algorithm, reducing memory overhead and runtime. (*Advisor: Bill Noble*)

TEACHING EXPERIENCE

Harvard University | Teaching Fellow, CS 2790R: Human-Computer Interaction

Fall 2025

- Facilitated student learning in graduate-level HCI concepts under Prof. Elena Glassman, holding weekly office hours and providing comprehensive feedback on project designs and implementations.

Harvard University | Teaching Fellow, APMTH 231: Decision Theory

Spring 2026

- Evaluated advanced coursework and provided technical feedback on core decision theory and statistical methodologies under the instruction of Prof. Demba Ba.

INVITED TALKS

- [1] **Traveling Waves Integrate Spatial Information Through Time** CCN 2025 (Aug 2025)
Neural Computations: Dynamics Across Space, Time, and Task
- [2] **Can Your Neurons Hear the Shape of an Object?** Kempner Institute (June 2025)
Frontiers in NeuroAI Symposium

AWARDS & FELLOWSHIPS

Kempner Institute Graduate Fellowship

2023 – Present

Shanahan Foundation Predoctoral Fellowship

2022

Burkhardt Family Endowed Scholarship

2020, 2021

Gary A. Kildall Endowed Scholarship

2019

Boeing Scholarship

2018

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, Java